STUDENT PROJECT DEVELOPMENT GUIDE

Python · Machine Learning · Natural Language Processing

Using AI Tools: Claude, GitHub Copilot, Hugging Face & More

Sample Project: Customer Review Sentiment Analyzer

A beginner-friendly, step-by-step walkthrough on how to plan, design, and build a real-world Python ML/NLP project using AI tools — without writing code from memory.

Note: This guide uses 'Customer Review Sentiment Analyzer' as a sample ML/NLP project to demonstrate how AI tools are used in a real data science workflow. You can apply these same steps to any Python ML or NLP project assigned to you.

What You Will Learn

By following this guide, you will be able to:

- Understand an ML/NLP project requirement and break it into tasks
- Use Claude to plan your project, understand algorithms, and write prompts
- Use GitHub Copilot in VS Code to generate Python code faster
- Use Hugging Face and other AI tools to load pre-trained models
- Train, evaluate, and improve a model step by step
- Push your project to GitHub and submit it professionally

AI Tools You Will Use

Tool	Best Used For	Where to Access
Claude (by Anthropic)	Understand requirements, plan architecture, explain ML concepts, generate prompts, debug logic	claude.ai (free account)
GitHub Copilot	AI code suggestions directly in VS Code as you type Python	VS Code + GitHub Student Pack (free)
Hugging Face	Access pre-trained NLP models (BERT, DistilBERT, etc.) and datasets	huggingface.co (free account)
ChatGPT / Gemini	Alternative for concept explanations or generating dataset ideas	chat.openai.com / gemini.google.com
Perplexity AI	Quick research on ML techniques, libraries, and paper summaries	perplexity.ai (free)
VS Code	Code editor for writing and running your Python project	code.visualstudio.com
GitHub	Store, version-control, and submit your project code	github.com

Google Colab	Run Python and ML code in the browser without any local setup	colab.research.google.com (free GPU)

Suggested Python ML / NLP Project Ideas

Pick any project from this list that is NOT already in your team assignment sheet:

Project	Type	Difficulty
Customer Review Sentiment Analyzer (Sample)	NLP — Classification	Beginner
Fake News Detection using Text Classification	NLP — Classification	Beginner
Resume Keyword Extractor and Job Match Scorer	NLP — Extraction	Intermediate
Medical Symptom Checker using NLP	NLP — Information Retrieval	Intermediate
Spam Email Classifier	ML — Classification	Beginner
Student Performance Predictor using ML	ML — Regression	Beginner
Movie Recommendation System using Collaborative Filtering	ML — Recommendation	Intermediate
Chatbot for College FAQs using NLP	NLP — Conversational AI	Intermediate
Loan Default Prediction using Machine Learning	ML — Classification	Intermediate
News Article Topic Classifier (Multi-class)	NLP — Classification	Intermediate

STEP 1 Understand the Project Requirements

Before writing any code or prompts, you must clearly understand what you are building. Use Claude to break down your ML/NLP project requirements into simple parts.

What to Do

- Read your assigned project description carefully from your Google Sheet.
- Open claude.ai and start a new conversation.
- Paste your project description and ask Claude to explain it simply.

Claude Prompt — Understanding Your ML/NLP Project

```
I am a student and I have been assigned the following project:  "[Paste
your project description here]"  Please: 1. Explain what this project is
in simple terms — what problem does it solve? 2. What kind of Machine
```

```
Learning or NLP task is this? (e.g. classification, regression,
clustering, text generation) 3. What Python libraries will I likely need?
(e.g. scikit-learn, pandas, NLTK, Hugging Face Transformers) 4. What
data/dataset will I need to complete this project? 5. Break the project
into 8-10 smaller tasks I can work on one by one.
```

Tip: Copy Claude's response into a notes file or Google Doc. This becomes your personal project plan. Label it 'Project Plan — [Your Project Name]'.

STEP 2 Set Up Your Development Environment

Before writing any code, set up all the tools you need so your Python environment is ready.

Install & Configure

- Install Python 3.10+ from python.org (if not already installed)
- Install VS Code from code.visualstudio.com
- Install the Python extension and GitHub Copilot extension in VS Code
- Sign in to GitHub Copilot using your student account (free via GitHub Education)
- Create a new project folder on your computer (e.g. sentiment-analyzer)
- Open that folder in VS Code and create a virtual environment

Claude Prompt — Environment Setup

```
I am starting a Python ML/NLP project called "[Your Project Name]".
Please give me: 1. The exact folder structure I should create for this
project 2. A list of all Python libraries I need to install (with pip
install commands) 3. What each library does in simple terms 4. How to
create and activate a virtual environment on Windows / Mac 5. What the
main Python files in my project should be named and what each one does
```

Tip: Use Google Colab (colab.research.google.com) if you cannot install Python locally. It runs entirely in your browser with free GPU access — perfect for ML projects.

Claude Prompt — Hugging Face Setup (for NLP Projects)

```
I am building an NLP project and want to use a pre-trained model from
Hugging Face.  Please: 1. Explain what Hugging Face is and why it is
useful for NLP 2. Recommend the best pre-trained model for [your task,
e.g. sentiment analysis / text classification] 3. Show me how to write a
prompt to load this model using the Hugging Face Transformers library 4.
Explain what a pipeline is in Hugging Face and when I should use it
```

STEP 3 Find and Understand Your Dataset

Every ML/NLP project needs data. Finding the right dataset and understanding its structure is one of the most important steps.

Claude Prompt — Finding a Dataset

```
I am building a [your project type, e.g. sentiment analysis] Python
project.  Please: 1. Recommend 3 free, beginner-friendly datasets I can
use from Kaggle, Hugging Face Datasets, or UCI ML Repository 2. For each
dataset, explain: what it contains, how many rows/columns, and why it
suits my project 3. Explain what format the data is usually in (CSV, JSON,
etc.) and how to load it with pandas 4. What columns/features should I
look for in a good dataset for this task?
```

Understanding Your Data

Claude Prompt — Understanding Your Dataset

```
I have downloaded a dataset for my [project type] project. Here is a
sample of the first 5 rows:  [Paste your dataset sample here — copy from
Excel or CSV preview]  Please: 1. Explain what each column means in simple
terms 2. Identify which column is the target/label I need to predict 3.
Identify which columns are the features (inputs) I will use 4. Tell me if
there is anything unusual or problematic in this data I should be aware of
```

Important: Never use a dataset with personal or sensitive real-world data (names, phone numbers, ID numbers) in your student project. Use publicly available, anonymized datasets from Kaggle or Hugging Face.

STEP 4 Plan Your Project Architecture with Claude

Before writing or generating any code, ask Claude to help you design the overall structure of your ML pipeline.

Claude Prompt — ML Pipeline Planning

```
I am building a [project name] using Python.  Dataset: [brief description
of your dataset] Goal: [what you want to predict or classify]  Please: 1.
Draw a simple text-based diagram showing the full ML pipeline from raw
data to final output 2. List each stage of the pipeline: data loading →
cleaning → preprocessing → model training → evaluation → output 3.
Describe what I should do at each stage in plain English 4. Recommend the
best ML algorithm for this task and explain why (e.g. Logistic Regression,
Naive Bayes, Random Forest, BERT) 5. List the Python functions I will need
to write and what each one should do
```

✓ **Tip:** Ask Claude 'What is the difference between [Algorithm A] and [Algorithm B] for my task?' to help you choose the right model. Understanding WHY you choose an algorithm is important for your project presentation.

STEP 5 Clean and Preprocess Your Data

Raw data is almost never ready to use directly. Data cleaning and preprocessing is where you prepare your data for the ML model.

Claude Prompt — Data Cleaning

```
My dataset for [project name] has the following issues I noticed:
[Describe the problems, e.g.: - Some rows have missing values in the
'review_text' column - The 'rating' column has some typos like '5.0' and
'5' - There are duplicate rows]  Please: 1. Explain what each problem
means and why it matters for ML 2. Describe (in plain English, no code
yet) what steps I need to take to fix each issue 3. What order should I
perform these cleaning steps in? 4. After cleaning, what should my data
look like?
```

For NLP Projects — Text Preprocessing

Claude Prompt — NLP Text Preprocessing

```
I am building an NLP project that works with raw text data.  My text
column contains: [describe your text, e.g. product reviews, tweets, news
articles]  Please explain (in plain English) what each of these
preprocessing steps does and whether I need it for my project: 1.
Lowercasing 2. Removing punctuation and special characters 3. Removing
stopwords (e.g. 'the', 'is', 'and') 4. Stemming vs Lemmatization — which
should I use? 5. Tokenization 6. Vectorization — TF-IDF vs Word Embeddings
— which is better for my task?
```

Copilot Prompt (in VS Code Comment)

```
# Function to preprocess raw text: lowercase, remove punctuation, remove
stopwords, lemmatize
```

✓ **Tip:** Use GitHub Copilot in VS Code by typing a descriptive comment above where you want the code, then press Enter. Copilot will suggest the function automatically. Press Tab to accept, or keep typing to get a different suggestion.

STEP 6 Train Your Model

This is the core step of your project — choosing and training the right machine learning model on your cleaned data.

Claude Prompt — Model Training Plan

```
I am ready to train my ML model for [project name].  My task is: [e.g.
binary sentiment classification — positive or negative] My dataset has [X]
rows and [Y] features/columns after preprocessing. I have chosen to use:
[your algorithm, e.g. Logistic Regression / DistilBERT]  Please: 1.
Explain how [your algorithm] works in simple terms — I am a beginner 2.
What does 'training' the model actually mean in plain English? 3. What is
a train/test split and what ratio should I use? (e.g. 80/20) 4. What
Python libraries and functions do I need for this step? 5. What should I
watch out for when training? (e.g. overfitting, data imbalance)
```

For NLP Projects Using Hugging Face

Claude Prompt — Hugging Face Fine-tuning

```
I want to fine-tune a pre-trained Hugging Face model for my [project task]
project.  Pre-trained model I want to use: [e.g. distilbert-base-uncased]
Please: 1. What is fine-tuning and how is it different from training a
model from scratch? 2. Should I fine-tune for this task or just use the
model's pipeline directly? Explain why. 3. What are the training arguments
I need to configure (epochs, learning rate, batch size)? 4. What does each
training argument do in simple terms? 5. How long will training
approximately take on Google Colab?
```

Important: Always read the code that Claude or Copilot generates. Do not copy-paste without understanding it. Ask Claude to explain any line you do not understand — type: 'Explain this line of code to me: [paste line]'.

STEP 7 Evaluate Your Model

After training, you must measure how well your model actually works. This is called evaluation. Never skip this step.

Claude Prompt — Understanding Evaluation Metrics

```
My ML model for [project name] has finished training. I need to evaluate
its performance.  My task type is: [e.g. binary classification / multi-
class classification / regression]  Please: 1. What evaluation metrics
should I use for this type of task? (e.g. Accuracy, Precision, Recall, F1,
RMSE, AUC-ROC) 2. Explain what each metric means in simple English — what
does a high/low score mean? 3. What is a Confusion Matrix and how do I
read it? 4. What is considered a 'good' score for each metric for my type
of project? 5. If my accuracy is high but F1 is low, what does that mean?
```

Evaluation Checklist

- Run predictions on your test set (data the model has never seen)
- Calculate accuracy, precision, recall, and F1 score
- Plot a confusion matrix to see where your model makes mistakes
- Identify which class/category the model struggles with most
- Ask Claude to interpret your results and suggest improvements

Claude Prompt — Interpreting Your Results

```
My [project name] model produced the following evaluation results:
Accuracy: [X]% Precision: [X] Recall: [X] F1 Score: [X] Confusion Matrix:
[paste values]  Please: 1. Tell me in plain English whether these results
are good, average, or poor for this type of task 2. What is the most
common error my model is making based on the confusion matrix? 3. What are
3 concrete things I can try to improve these scores? 4. Should I try a
different algorithm? Which one and why?
```

STEP 8 Debug Problems with AI Assistance

Errors and bugs are a normal part of every project. Use AI tools to understand and fix them — not to panic.

Claude Prompt — Debugging Errors

```
My Python ML project has an error. Here is the full error message:  [Paste
the complete error traceback from your terminal/console here]  Here is the
relevant part of my code:  [Paste the Python function or code block that
is causing the error]  Please: 1. Identify exactly what is causing this
error 2. Explain why it is happening in simple terms 3. Show me the
corrected approach 4. Explain the fix so I can learn from it and avoid it
next time
```

Common ML/NLP Errors and What to Ask

Error / Problem	Ask Claude or Copilot
ModuleNotFoundError: No module named 'X'	'How do I install [library] for Python? I am using pip inside a virtual environment.'
ValueError: could not convert string to float	'My ML model is getting a ValueError on my text column. How do I fix string data for sklearn?'
Model accuracy is 50% — same as random guessing	'My classifier has 50% accuracy on a binary dataset. Why might this happen and how do I fix it?'
MemoryError when loading large dataset	'I am getting a MemoryError loading my CSV. How do I load a large dataset in chunks using pandas?'
Overfitting — high train accuracy, low test accuracy	'My model has 97% train accuracy but 60% test accuracy. What is overfitting and how do I fix it?'

CUDA / GPU errors in Google Colab	'I am getting a CUDA error in Google Colab when running a Hugging Face model. How do I fix this?'

✅ **Tip:** In VS Code, open the integrated terminal and copy the exact red error text. Paste it into Claude with a brief description of what you were trying to do. The more context you give, the better Claude can help.

STEP 9 Push Your Project to GitHub

GitHub is where you store and submit your code. Push your project after every major step — not just at the end.

Steps to Upload Your Project

- Go to github.com and create a New Repository
- Name it the same as your project (e.g. sentiment-analyzer)
- Set it to Public so your instructor can view it
- Open VS Code Terminal (Ctrl + ` or Terminal → New Terminal)
- Run the Git commands shown in the table below:

Command	What It Does
git init	Starts tracking your project folder with Git
git add .	Stages all your files to be saved (excludes files in .gitignore)
git commit -m "Initial commit"	Saves a snapshot of your project with a message
git branch -M main	Names your main branch 'main'
git remote add origin [your-repo-url]	Links your local folder to your GitHub repository
git push -u origin main	Uploads your code to GitHub for the first time
git add . && git commit -m "message" && git push	Quick command to save and upload future changes

Note: Add a .gitignore file to your project to avoid uploading your virtual environment folder (venv/) and large dataset files. Ask Claude: 'Write a .gitignore file for a Python ML project using a virtual environment and storing data in a /data folder.'

Claude Prompt — Git Help

```
I am trying to push my Python project to GitHub and I am getting this
error:  [Paste the exact error message here]  Please explain what this
error means and give me the exact steps to fix it.
```

STEP 10 Write a README and Project Report

A README explains your project to anyone who views it on GitHub. A project report summarizes your methodology, results, and learning. Both are professional practices that improve your submission quality.

Claude Prompt — README Generation

```
Write a professional README.md file for my GitHub project.  Project name: [Your project name] Description: [Brief description of what it does] Type: Python ML / NLP project Libraries used: [e.g. scikit-learn, pandas, NLTK, Hugging Face Transformers] Dataset: [Dataset name and source] Model used: [e.g. Logistic Regression / DistilBERT] Results: [Your best accuracy/F1 score]  Include sections for: - Project Description - Problem Statement - Dataset Used - Methodology (steps taken) - Results and Evaluation - How to Run the Project - Technologies Used - Screenshots or output examples placeholder
```

Claude Prompt — Project Report Help

```
Help me write a project report for my [project name] ML/NLP project.  Here is a summary of what I did: - Dataset: [name, size, source] - Preprocessing: [steps taken] - Model: [algorithm used] - Results: Accuracy [X]%, F1 Score [X]  Please write: 1. A 2-paragraph introduction explaining the problem and why it matters 2. A methodology section describing each step I took 3. A results section interpreting my scores 4. A conclusion section with what I learned and what I would improve next time
```

STEP 11 Review, Improve & Submit

Before submitting, review your work and use AI to help you improve code quality, clarity, and completeness.

Claude Prompt — Code Review

```
Please review my complete Python ML project code below.  [Paste your main Python script(s) here]  Please: 1. Identify any bugs, logical errors, or missing steps 2. Suggest improvements for code readability (naming, structure, comments) 3. Are there any bad practices I should fix? 4. Is my evaluation done correctly? 5. Suggest one additional feature or improvement I could add to make the project more impressive
```

Final Submission Checklist

- All features and requirements from the project description are implemented
- Data cleaning and preprocessing are clearly done and explained in comments

- Model is trained, evaluated, and results are printed or saved
- Evaluation metrics (accuracy, F1, confusion matrix) are included
- README.md is written and added to the GitHub repository
- All Python files have comments explaining each section
- Project is pushed to GitHub and the repository is set to Public
- GitHub repository link is submitted to the instructor
- You can explain every part of your project in your own words

Quick Prompt Reference Card

Save these prompts. Use them any time you are stuck during development.

Situation	What to Ask Claude or Copilot
I don't understand a concept	"Explain [concept, e.g. F1 Score / BERT / TF-IDF] to me like I am a beginner. Give a simple real-world example."
I have a Python error	"My code has this error: [paste traceback]. Here is my code: [paste code]. Please fix it and explain why it happened."
I want to understand code	"Explain this Python code line by line: [paste code]"
I am stuck on logic	"I need to write a Python function that does [describe task]. What approach should I take and what library should I use?"
I want to improve my code	"Review this Python code and suggest improvements for readability, efficiency, and ML best practices: [paste code]"
I need to choose an algorithm	"I am doing [task type] with [dataset description]. Should I use [Algorithm A] or [Algorithm B]? Explain the tradeoffs."
My model is not performing well	"My model's F1 score is [X]. Here are my preprocessing steps and algorithm: [describe]. What can I do to improve performance?"
I need a new feature	"How do I add [feature, e.g. confidence score display] to my existing Python ML pipeline? Describe the approach step by step."
I have a Hugging Face question	"I am using the [model name] from Hugging Face for [task]. I am confused about [specific issue]. Please explain."
I get a Git error	"I ran git push and got this error: [paste error]. What does it mean and how do I fix it?"

Best AI Tool for Each Project Stage

Project Stage	Best AI Tool	Why
Understanding requirements & planning	Claude	Best at explaining concepts, breaking down problems, and creating step-by-step plans
Writing Python code	GitHub Copilot (in VS Code)	Generates code inline as you type — fastest for coding tasks
Finding pre-trained NLP models	Hugging Face	World's largest repository of open-source ML models and datasets
Finding and exploring datasets	Kaggle + Hugging Face Datasets	Free, labeled datasets ready for ML projects with community notebooks
Debugging errors	Claude or ChatGPT	Excellent at reading error tracebacks and explaining root causes
Researching ML algorithms	Perplexity AI	Gives research-backed answers with source links — great for theory questions

Running ML code (no local setup)	Google Colab	Free browser-based Python notebook with GPU access — ideal for training
Understanding papers or techniques	Claude or Perplexity AI	Can summarize complex research in beginner-friendly language
Code review and improvement	Claude	Analyzes entire code blocks and gives structured, constructive feedback
Version control and submission	GitHub + VS Code	Industry-standard tools for code management and collaboration

Important Do's and Don'ts

✅ DO	❌ DON'T
Ask Claude to explain any code it generates	Copy code without reading or understanding it
Use AI as a learning and planning assistant	Use AI to do your entire project without effort
Test your model after every major change	Train and submit without checking your evaluation metrics
Ask follow-up questions if you are confused	Submit a project you cannot explain in your own words
Add comments to every function explaining what it does	Leave code uncommented and messy
Use Claude to understand WHY an algorithm works	Just guess or randomly pick a model without reasoning
Push code to GitHub after every major step	Wait until the last minute to upload everything
Use publicly available, anonymized datasets	Use datasets with personal or sensitive private data

Final Message to Students

AI tools like Claude, GitHub Copilot, and Hugging Face are your learning partners — not shortcuts. The goal is to understand how to build things, not just get a final output. Use every prompt as a learning opportunity. Ask "why" and "how" at every step. Machine Learning and NLP are powerful, real-world skills. The understanding you build now will stay with you throughout your career.

Good luck with your project! You have got this.